Project Initialization and Planning Phase

Date	13 July 2026
Team ID	
Project Title	Smart Lender – Loan Eligibility Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to assist farmers in selecting the most suitable crop using machine learning. The system analyzes soil nutrients and environmental conditions to recommend the best crop, enabling informed farming decisions, improving productivity, and reducing the risk of poor crop selection. The application provides a simple web interface for users to enter agricultural parameters and receive instant recommendations.

Project Overview	
Objective	Develop an intelligent crop recommendation system that predicts the most suitable crop based on Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, and Rainfall using machine learning.
Scope	The project provides AI-based crop recommendations through a web application. It supports farmers by analyzing soil and weather conditions and suggesting the most appropriate crop for cultivation.
Problem Statement	
Description	Farmers often face challenges in selecting suitable crops because of varying soil nutrients and environmental conditions. Traditional decision-making methods may not always produce optimal results.
Impact	Incorrect crop selection can reduce agricultural productivity, increase cultivation costs, and decrease farmers' income. A data-driven recommendation system helps improve decision-making and farming outcomes.
Proposed Solution	
Approach	Develop a Flask-based web application integrated with a trained machine learning model. The system accepts soil and weather parameters, processes them, and predicts the most suitable crop.
Key Features	- Input using N, P, K, Temperature, Humidity, pH, and Rainfall.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv